

BVAR Scenario Tool

User Manual and Methodological Documentation¹

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International Monetary Fund

June 7, 2026

Source code: <https://github.com/matyasfarkas/BVARScenarioTool>

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Abstract

This document describes the BVAR Scenario Tool, a browser-based application for conditional forecasting, scenario analysis, and policy counterfactual evaluation. The tool combines a large Bayesian vector autoregression (BVAR) estimated with hierarchical Minnesota priors ([Giannone et al., 2015](#)) with structural impulse responses drawn from the Macroeconomic Model Data Base (MMB; [Wieland et al., 2012, 2016](#)). Users condition the BVAR forecast on arbitrary variable paths via a Durbin–Koopman simulation smoother ([Durbin and Koopman, 2002, 2012](#)), and evaluate monetary and fiscal policy counterfactuals using DSGE-identified impulse response functions. We describe the methodology employed, the implementation of the conditional forecasting algorithms, and the three-layer architecture that separates the unconditional baseline, the user’s scenario, and the policy counterfactual. All computation runs client-side in the browser; the tool requires no server infrastructure and can be deployed as a static website.

Keywords: BVARs, Conditional Forecasting, Scenario Analysis, Kalman Filter, Durbin–Koopman Smoother, Policy Counterfactuals, DSGE Models, MMB

JEL classification: C32, C53, E37, E52, E58

*The views in this paper are solely the responsibility of the author and should not be interpreted as reflecting the views of the International Monetary Fund or any other person associated with the IMF. Email: mfarkas@imf.org.

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1 Introduction

Conditional forecasting with Bayesian vector autoregressions (BVARs) has become a standard tool in central bank forecasting and policy analysis. The seminal contributions of [Doan et al. \(1984\)](#) and [Waggoner and Zha \(1999\)](#) established the methodological foundations, while [Bańbura et al. \(2015\)](#) extended these techniques to large cross-sections using state-space methods. The BVAR Scenario Tool described in this document builds on this literature to provide an interactive, browser-based platform that makes conditional forecasting and policy counterfactual analysis accessible without specialized software.

The tool implements a three-layer analytical framework:

- (i) **Baseline forecast:** An unconditional forecast from a large BVAR estimated on quarterly macroeconomic and financial data, following the hierarchical prior framework of [Giannone et al. \(2015\)](#).
- (ii) **Conditional scenario:** The user conditions any subset of variables on desired paths; a Durbin–Koopman smoother ([Durbin and Koopman, 2002, 2012](#)) propagates these constraints to all remaining variables, respecting the full covariance structure of the VAR.
- (iii) **Policy counterfactual:** An alternative monetary or fiscal policy path is overlaid on the scenario using impulse response functions from a DSGE model drawn from the Macroeconomic Model Data Base (MMB; [Wieland et al., 2012](#)), and the resulting macroeconomic responses are re-conditioned through the BVAR to ensure cross-variable consistency.

The tool is designed for economists at central banks and international institutions who need to construct and communicate conditional projections and policy scenarios. It runs entirely in the browser, requiring no server-side computation, and can be deployed as a static website on any hosting platform.

This document is organized as follows. Section 2 describes the BVAR estimation methodology. Section 3 derives the companion-form state-space representation. Section 4 presents the Kalman filter and Durbin–Koopman smoother used for conditional forecasting. Section 5 details the policy counterfactual framework. Section 6 documents the variable coverage. Section 7 provides a guide to the user interface. Section 8 discusses technical implementation details. Section 9 concludes with a discussion of limitations and directions for future development.

2 BVAR Estimation

2.1 Reduced-Form VAR

Consider an n -variable VAR(p) in reduced form:

$$y_t = c + \sum_{\ell=1}^p B_{\ell} y_{t-\ell} + u_t, \quad u_t \sim \mathcal{N}(0, \Sigma), \quad (1)$$

where $y_t \in \mathbb{R}^n$ is the vector of observables, $c \in \mathbb{R}^n$ is a vector of intercepts, $B_1, \dots, B_p \in \mathbb{R}^{n \times n}$ are the autoregressive coefficient matrices, and $\Sigma \in \mathbb{R}^{n \times n}$ is the positive-definite innovation covariance matrix. In the US specification, $n = 36$ and $p = 3$, yielding a coefficient matrix $\hat{\beta} \in \mathbb{R}^{109 \times 36}$ (with row 0 containing the intercept).

2.2 Data Transformations

Variables enter the VAR in one of two forms:

- **Log-transformed** (`log`): the variable is modeled in log levels. Growth rates displayed to the user are quarter-over-quarter annualized:

$$g_t^{\text{q/q ann.}} = 400 \times (\ln Y_t - \ln Y_{t-1}). \quad (2)$$

Year-over-year growth rates are computed as:

$$g_t^{\text{y/y}} = 100 \times (\ln Y_t - \ln Y_{t-4}). \quad (3)$$

- **Linear** (`lin`): the variable enters in levels (e.g., interest rates in percent, unemployment rate). No growth-rate conversion is applied.

2.3 Reporting Units

The display unit is chosen per variable for interpretability, independently of the estimation transformation:

- **National-accounts / activity flows** (GDP, consumption, investment, trade volumes) are shown as growth (q/q annualized or y/y).
- **Prices** (CPI/HICP/deflators) are shown as inflation.
- **Rates and ratios** (policy and market rates, unemployment, fiscal ratios) are shown as levels in percent / percentage points. Unemployment-rate uncertainty bands are floored at 0 (a rate cannot be negative); policy and market rates are *not* floored, because negative nominal rates do occur.
- **Index / FX / equity / commodity series** (the trade-weighted dollar FXTWM, SP500, house prices HP, oil PZTEXP, the non-fuel commodity index GSNE, and the other blocs' NEER_EA, NEER_UK, NEER_CA, REER_CN, EQ_CN) are reported as the *index level* (e.g. 2010 = 100) rather than as q/q growth: a trade-weighted exchange rate or equity index q/q-annualized is not an informative number, whereas its level is. These series remain `log` for estimation and are conditioned in the stored $100 \ln(\text{index})$ space; only the display is exponentiated back to the level, $\exp(\text{stored}/100)$, with a level-space uncertainty fan.

2.4 Hierarchical Minnesota Prior

The model is estimated following [Giannone et al. \(2015\)](#), who cast the Minnesota prior ([Litterman, 1986](#); [Doan et al., 1984](#)) in a hierarchical framework with data-driven hyperparameter selection. The prior specification includes:

- (a) The **Minnesota shrinkage** prior on B_ℓ , centered on the random-walk or white-noise prior for each variable (depending on the variable's stationarity properties). The tightness parameter λ controls the overall degree of shrinkage toward the prior mean.
- (b) The **sum-of-coefficients** prior ([Doan et al., 1984](#)), which shrinks toward the belief that a unit increase in every variable's level produces a unit increase in its forecast, imposing a form of cointegration.
- (c) The **single-unit-root** prior, which constrains the system to have at most one common stochastic trend.

The hyperparameters governing the tightness of each prior component are treated as additional parameters and selected by maximizing the marginal likelihood, as advocated by [Giannone et al. \(2015\)](#).

Posterior draws are obtained via Markov chain Monte Carlo (MCMC). The tool uses 5,000 draws with a burn-in of 2,500, retaining the posterior mean $\hat{\beta}$ and $\hat{\Sigma}$ for the deterministic conditional forecasting engine, and storing the full set of 2,500 posterior draws $\{(\hat{\beta}^{(d)}, \hat{\Sigma}^{(d)})\}_{d=1}^D$ for uncertainty quantification via the simulation smoother (Section 4.6).

3 Companion-Form State Space

To apply the Kalman filter, the VAR(p) in (1) is cast in companion form as a first-order state-space model (see, e.g., [Durbin and Koopman, 2012](#), ch. 3).

3.1 State Vector

Define the np -dimensional state vector:

$$\alpha_t = \begin{pmatrix} y_t \\ y_{t-1} \\ \vdots \\ y_{t-p+1} \end{pmatrix} \in \mathbb{R}^{np}. \quad (4)$$

3.2 Transition Equation

The state evolves according to:

$$\alpha_t = T \alpha_{t-1} + d + R \eta_t, \quad \eta_t \sim \mathcal{N}(0, \Sigma), \quad (5)$$

where the $np \times np$ transition matrix T , the np -dimensional intercept vector d , and the $np \times n$ selection matrix R are:

$$T = \begin{pmatrix} B'_1 & B'_2 & \cdots & B'_{p-1} & B'_p \\ I_n & 0 & \cdots & 0 & 0 \\ 0 & I_n & \cdots & 0 & 0 \\ \vdots & & \ddots & & \vdots \\ 0 & 0 & \cdots & I_n & 0 \end{pmatrix}, \quad d = \begin{pmatrix} c \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad R = \begin{pmatrix} I_n \\ 0 \\ \vdots \\ 0 \end{pmatrix}. \quad (6)$$

Remark 1. The B'_ℓ blocks in T are transposed relative to (1) because the coefficient matrix $\hat{\beta}$ is stored as $(1 + np) \times n$, with row 0 containing the intercept c and rows $1, \dots, np$ containing the stacked lag coefficients. Specifically, $T(i, j) = \hat{\beta}(j + 1, i)$ for $0 \leq i < n$ and $0 \leq j < np$.

3.3 Observation Equation

The observation equation extracts the contemporaneous variables:

$$y_t = Z \alpha_t + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, H), \quad (7)$$

where $Z = (I_n \mid 0_{n \times n(p-1)})$ and $H = \sigma_\varepsilon^2 I_n$ with $\sigma_\varepsilon^2 = 10^{-12}$ (a negligible measurement error included for numerical stability of the Kalman filter recursions).

3.4 State-Noise Covariance

The $np \times np$ state-noise covariance matrix is:

$$Q = R \Sigma R' = \begin{pmatrix} \Sigma & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & & \ddots & \\ 0 & 0 & \cdots & 0 \end{pmatrix}. \quad (8)$$

4 Kalman Filter and Smoother

4.1 Conditioning via Partial Observations

The key insight underlying conditional forecasting in a state-space framework, following Bańbura et al. (2015), is that conditioning on variable i at horizon h is equivalent to observing $y_{T+h,i}$ in the Kalman filter. For each forecast period $t = T + 1, \dots, T + H$, the observation vector is partitioned into:

- **Conditioned variables:** $y_t^{(o)}$, with known target values specified by the user.
- **Free variables:** $y_t^{(m)}$, treated as missing observations (coded as NaN).

The Kalman filter processes only the observed subset at each time step, adapting Z_t , H_t , and the innovation v_t accordingly.

4.2 Forward Pass: Kalman Filter

Algorithm: Kalman Filter with Partial Observations. **Input:** T, Z, Q, H, d ; initial state α_0 , covariance P_0 ; partial observations $\{y_t^{(o)}\}_{t=1}^H$.

For $t = 1, \dots, H$:

$$\alpha_{t|t-1} = T \alpha_{t-1|t-1} + d \quad (\text{state prediction}) \quad (9)$$

$$P_{t|t-1} = T P_{t-1|t-1} T' + Q \quad (\text{covariance prediction}) \quad (10)$$

If observations exist at time t , let $Z_t^{(o)}$ be the rows of Z for observed variables, then:

$$v_t = y_t^{(o)} - Z_t^{(o)} \alpha_{t|t-1} \quad (\text{innovation}) \quad (11)$$

$$F_t = Z_t^{(o)} P_{t|t-1} (Z_t^{(o)})' + H_t^{(o)} \quad (\text{innovation variance}) \quad (12)$$

$$K_t = P_{t|t-1} (Z_t^{(o)})' F_t^{-1} \quad (\text{Kalman gain}) \quad (13)$$

$$\alpha_{t|t} = \alpha_{t|t-1} + K_t v_t \quad (\text{state update}) \quad (14)$$

$$P_{t|t} = P_{t|t-1} - K_t Z_t^{(o)} P_{t|t-1} \quad (\text{covariance update}) \quad (15)$$

If no observations exist at time t : $\alpha_{t|t} = \alpha_{t|t-1}$ and $P_{t|t} = P_{t|t-1}$.

The matrix inversion F_t^{-1} is computed via LU decomposition with partial pivoting; singularity is flagged when the pivot falls below 10^{-15} .

4.3 Backward Pass: Durbin–Koopman Smoother

The smoother refines the filtered estimates using all future information, yielding $\alpha_{t|H}$ for all t ([Durbin and Koopman, 2002, 2012](#)).

Algorithm: Durbin–Koopman Fixed-Interval Smoother. **Input:** Filtered quantities $\{\alpha_{t|t-1}, P_{t|t-1}, v_t, Z_t^{(o)}\}$, Initialize $r_H = 0$. For $t = H, H-1, \dots, 1$:

If observations exist at time t :

$$L_t = T' (I - P_{t|t-1} (Z_t^{(o)})' F_t^{-1} Z_t^{(o)}), \quad (16)$$

$$r_{t-1} = (Z_t^{(o)})' F_t^{-1} v_t + L_t' r_t. \quad (17)$$

Otherwise: $r_{t-1} = T' r_t$.

The smoothed state is:

$$\alpha_{t|H} = \alpha_{t|t-1} + P_{t|t-1} r_{t-1}. \quad (18)$$

4.4 Conditional Forecast Computation

The conditional forecast is computed as follows:

1. Initialize the state from the last p historical observations: $\alpha_0 = (y'_T, y'_{T-1}, \dots, y'_{T-p+1})'$.
2. Construct the conditioning matrix $Y^* \in \mathbb{R}^{H \times n}$, where $Y_{t,j}^*$ is the target value for variable j at horizon t if conditioned, and **NaN** otherwise.
3. Run the Kalman filter (Algorithm 4.2) and smoother (Algorithm 4.3) with partial observations given by the non-**NaN** entries of Y^* .
4. Extract the observation block (first n elements) from each smoothed state $\alpha_{t|H}$.

4.5 Delta-Based Display

To ensure that zero user deviations produce exactly the stored baseline forecast, the tool computes both an unconditional smoother pass (with all entries of Y^* set to **NaN**) and a conditional pass. The displayed forecast is:

$$\hat{y}_{T+h}^{\text{display}} = \bar{y}_{T+h}^{\text{baseline}} + \underbrace{(\hat{y}_{T+h}^{\text{cond}} - \hat{y}_{T+h}^{\text{uncond}})}_{\text{smoother delta}}, \quad (19)$$

where $\bar{y}^{\text{baseline}}$ is the pre-computed baseline forecast (posterior mean). This construction guarantees that the displayed forecast matches the stored baseline exactly when no conditioning constraints are imposed, avoiding small numerical discrepancies between the smoother's unconditional output and the stored baseline. The unconditional smoother result is cached and reused across conditioning changes.

For log-transformed variables, the smoother operates in log-level space, and the delta is scaled to annualized growth-rate space:

$$\Delta g_h = 4 \times \Delta \ell_h, \quad \Delta \ell_h = \ell_h^{\text{cond}} - \ell_h^{\text{uncond}}, \quad (20)$$

where ℓ_h denotes the log-level forecast at horizon h .

4.6 Uncertainty Quantification

Forecast uncertainty bands are computed via the Durbin–Koopman *simulation smoother* (Durbin and Koopman, 2002), which generates draws from the posterior predictive distribution conditional on the user's constraints.

For each posterior draw $d = 1, \dots, D$:

1. **Forward simulation:** Draw shocks $\eta_t^+ \sim \mathcal{N}(0, I_n)$ and compute a simulated path using the d -th posterior draw of the VAR coefficients:

$$\alpha_t^+ = T^{(d)} \alpha_{t-1}^+ + d^{(d)} + R L^{(d)} \eta_t^+, \quad (21)$$

where $L^{(d)}$ is the lower Cholesky factor of $\Sigma^{(d)}$. The simulated observations are $y_t^+ = Z \alpha_t^+$.

2. **Deviation computation:** At conditioned entries, compute $y_t^* = Y_t^{\text{target}} - y_t^+$; at free entries, set $y_t^* = \text{NaN}$.
3. **Smoother on deviations:** Run the Kalman smoother on $\{y_t^*\}$ with initial state $\alpha_0^* = 0$ and small initial covariance $P_0^* = 10^{-7} I_{np}$, producing adjustment paths $\hat{\alpha}_t^*$.
4. **Combine:** The conditional draw is $\tilde{\alpha}_t = \hat{\alpha}_t^* + \alpha_t^+$.

The 16th and 84th percentiles across D draws yield the reported 68% credible bands. The tool uses $D = 50$ in “Quick” mode and $D = 200$ in “Full” mode. The simulation smoother correctly integrates over both parameter uncertainty (through the posterior draws) and forecast uncertainty (through the simulated shocks).

5 Policy Counterfactuals

The counterfactual layer evaluates how macroeconomic outcomes change under an alternative policy stance. It operates *on top of* the conditional scenario described in Section 4: starting from the DK-smoothed scenario paths, the user specifies deviations in the policy rate or government spending, and the tool traces the implied effects through a DSGE model’s impulse response functions before re-conditioning the full system through the BVAR.

5.1 DSGE Impulse Response Functions

Impulse response functions (IRFs) are drawn from the Macroeconomic Model Data Base (MMB; [Wieland et al., 2012, 2016](#)). The MMB provides a library of pre-solved DSGE models with standardized variable names and multiple policy rule specifications. The tool includes IRFs from 169 models under 9 monetary policy rules:

Rule	Reference
Taylor	Taylor (1993)
SW	Smets and Wouters (2007)
CEE	Christiano et al. (2005)
GR	Gerdesmeier & Roffia (2004)
LWW	Levin, Wieland & Williams (2003)
OW08	Orphanides & Wieland (2008)
OW13	Orphanides & Wieland (2013)
Coenen	Coenen et al. (2012)
CMR	Christiano, Motto & Rostagno (2014)

Each IRF file provides the dynamic response of key macroeconomic variables—inflation, output gap, interest rate, consumption, investment, and others—to a one-standard-deviation monetary

policy shock under the specified rule. Fiscal policy IRFs are available for 60 models that include a fiscal sector.

5.2 Monetary Policy Shock Inversion

Given a user-specified interest-rate adjustment path $\Delta r = (\Delta r_1, \dots, \Delta r_H)'$, the tool solves for the structural monetary policy shock sequence $\varepsilon = (\varepsilon_1, \dots, \varepsilon_H)'$ via the lower-triangular Toeplitz system:

$$\underbrace{\begin{pmatrix} h_r(0) & 0 & \cdots & 0 \\ h_r(1) & h_r(0) & \cdots & 0 \\ \vdots & & \ddots & \vdots \\ h_r(H-1) & h_r(H-2) & \cdots & h_r(0) \end{pmatrix}}_{\mathcal{H}_r \in \mathbb{R}^{H \times H}} \varepsilon = \Delta r, \quad (22)$$

where $h_r(s)$ is the impulse response of the interest rate at horizon s . Since $\mathcal{H}_r$ is lower-triangular, the system is solved by forward substitution:

$$\varepsilon_t = \frac{\Delta r_t - \sum_{j=0}^{t-1} h_r(t-j) \varepsilon_j}{h_r(0)}, \quad t = 0, 1, \dots, H-1. \quad (23)$$

The implied responses of inflation and output are:

$$\Delta \pi_t^{\text{IRF}} = \sum_{j=0}^t h_\pi(t-j) \varepsilon_j, \quad \Delta y_t^{\text{IRF}} = \sum_{j=0}^t h_y(t-j) \varepsilon_j. \quad (24)$$

An analogous procedure applies to fiscal policy shocks, using the government-spending IRFs (h_g , h_π^g , h_y^g , h_r^g) from the DSGE models in the MMB that include a fiscal sector.

5.3 Three Conditioning Modes

After computing the DSGE-implied deltas, the counterfactual is re-conditioned through the BVAR to enforce cross-variable consistency. Three modes govern what is held fixed versus free in the second-stage DK smoother pass:

- (a) **Hold-targets:** All user-conditioned variables are pinned at their scenario values; the interest rate is pinned at the scenario rate plus the adjustment. Remaining variables are free and determined by the DK smoother:

$$Y_t^{\text{cf}}(j) = \hat{y}_{T+t,j}^{\text{cond}}, \quad \text{for } j \in \mathcal{C}, \quad (25)$$

$$Y_t^{\text{cf}}(r) = \hat{y}_{T+t,r}^{\text{cond}} + \Delta r_t, \quad (26)$$

$$Y_t^{\text{cf}}(j) = \text{NaN}, \quad \text{otherwise.} \quad (27)$$

- (b) **Let-adjust**: Only the policy instrument(s) are conditioned; all other variables, including inflation and output, are free and adjust endogenously through the BVAR covariance structure.
- (c) **IRF-consistent** (default): The interest rate is pinned at the scenario rate plus the adjustment, and inflation and output are pinned at the scenario values plus the DSGE-implied deltas from (24):

$$Y_t^{\text{cf}}(\pi) = \hat{y}_{T+t,\pi}^{\text{cond}} + \Delta \pi_t^{\text{IRF}}, \quad (28)$$

$$Y_t^{\text{cf}}(y) = \hat{y}_{T+t,y}^{\text{cond}} + \Delta y_t^{\text{IRF}}, \quad (29)$$

$$Y_t^{\text{cf}}(r) = \hat{y}_{T+t,r}^{\text{cond}} + \Delta r_t. \quad (30)$$

The remaining variables are determined by the DK smoother, producing responses consistent with both the DSGE-identified monetary transmission and the BVAR covariance structure.

In all modes, the counterfactual display uses the same delta construction as (19):

$$\hat{y}_{T+h}^{\text{cf,display}} = \hat{y}_{T+h}^{\text{scen,display}} + f(\hat{y}_{T+h}^{\text{cf,smooth}} - \hat{y}_{T+h}^{\text{scen,smooth}}), \quad (31)$$

where $f(\cdot)$ applies the $\times 4$ scaling from (20) for log-transformed variables.

5.4 Optimal Policy: Barnichon–Mesters

Following Barnichon and Mesters (2023), the tool solves for the optimal monetary policy shock sequence that minimizes a quadratic loss function over deviations from target:

$$\mathcal{L} = \sum_{t=1}^H \left[\lambda_{\pi} (\pi_t - \pi^*)^2 + \lambda_y (y_t - y^*)^2 \right], \quad (32)$$

subject to the linear mapping from shocks to outcomes given by the DSGE impulse responses.

Define the Toeplitz matrices $\mathcal{H}_{\pi}, \mathcal{H}_y \in \mathbb{R}^{H \times H}$ analogously to (22) using the inflation and output-gap IRFs. Stacking the weighted system:

$$\underbrace{\begin{pmatrix} \sqrt{\lambda_{\pi}} \mathcal{H}_{\pi} \\ \sqrt{\lambda_y} \mathcal{H}_y \end{pmatrix}}_{\tilde{\mathcal{H}} \in \mathbb{R}^{2H \times H}} \varepsilon = \underbrace{\begin{pmatrix} \sqrt{\lambda_{\pi}} \Delta \pi^{\text{target}} \\ \sqrt{\lambda_y} \Delta y^{\text{target}} \end{pmatrix}}_{\tilde{b} \in \mathbb{R}^{2H}}, \quad (33)$$

the optimal shock sequence is obtained via Tikhonov-regularized least squares:

$$\varepsilon^* = (\tilde{\mathcal{H}}' \tilde{\mathcal{H}} + \alpha I_H)^{-1} \tilde{\mathcal{H}}' \tilde{b}, \quad (34)$$

with regularization parameter $\alpha = 10^{-6}$. The implied policy rate path is $\Delta r_t^* = \sum_{j=0}^t h_r(t-j) \varepsilon_j^*$, which is then passed to the IRF-consistent conditioning mode (Section 5.3c) to obtain the full counterfactual.

The user controls the relative weight on price stability versus full employment via a slider that sets $\lambda_\pi/(\lambda_\pi + \lambda_y)$.

6 Variable Coverage

The US model includes $n = 36$ quarterly macroeconomic and financial variables with $p = 3$ lags. Table 1 lists the full specification.

Table 1: Variable coverage for the US BVAR model. “Trans.” denotes the data transformation: **log** = log levels (growth rates displayed), **lin** = levels. “Prior” denotes the prior mean: RW = random walk.

ID	Description	Trans.	Prior
GDPH	Real Gross Domestic Product	log	RW
CH	Real Personal Consumption Expenditures	log	RW
FRH	Real Private Residential Fixed Investment (nom/deflator)	log	RW
FNH	Real Private Nonresidential Fixed Investment (nom/deflator)	log	RW
XH	Real Exports of Goods & Services	log	RW
MH	Real Imports of Goods & Services	log	RW
GH	Real Government Consumption Expenditures & Gross Investment	log	RW
JGDP	Gross Domestic Product: Chain Price Index	log	RW
JC	Personal Consumption Expenditures: Chain Price Index	log	RW
JCXFE	PCE less Food & Energy: Chain Price Index	log	RW
UI	CPI-U: All Items	log	RW
UIXFDG	CPI-U: All Items Less Food & Energy	log	RW
LXBC	Business Sector: Compensation Per Hour (nominal)	log	RW
LANAGRA	All Employees: Total Nonfarm	log	RW
LR	Civilian Unemployment Rate: 16 yr +	lin	RW
IP	Industrial Production Index	log	RW
CUMFG	Capacity Utilization: Manufacturing [SIC]	lin	RW
HST	Housing Starts	log	RW

ID	Description	Trans.	Prior
YPDH	Real Disposable Personal Income	log	RW
CSENT	University of Michigan: Consumer Sentiment	lin	RW
POLRATE	Effective Federal Funds Rate (policy rate)	lin	RW
FCM2	2-Year Treasury Note Yield at Constant Maturity	lin	RW
FCM5	5-Year Treasury Note Yield at Constant Maturity	lin	RW
FCM10	10-Year Treasury Note Yield at Constant Maturity	lin	RW
FAAA	Moodys Seasoned Aaa Corporate Bond Yield	lin	RW
FBAA	Moodys Seasoned Baa Corporate Bond Yield	lin	RW
FXTWM	Nominal Trade-Weighted Exch Value of US\$ vs AFE Currencies	log	RW
SP500	Equity Price Index (OECD US share prices / S&P, public)	log	RW
PZTEXP	Spot Oil Price: West Texas Intermediate	log	RW
GSNE	Non-Fuel Commodity Price Index (IMF, non-energy proxy)	log	RW
SP500VOL	CBOE Volatility Index for S&P 500 Stock Price Index	lin	WN
HP	House Price	log	RW
LEV	Real nonfinancial business debt	log	RW
FREV_GDP	Federal Receipts (% of GDP)	lin	RW
FPEXP_GDP	Federal Primary Expenditure (% of GDP)	lin	RW
DDEBT_GDP	Change in Debt-to-GDP (pp)	lin	RW

Models for the five other economies—Euro Area ($n = 13$), Japan ($n = 13$), United Kingdom ($n = 15$), Canada ($n = 15$), and China ($n = 10$)—follow the same estimation methodology with country-appropriate variable selections. Each is a *Global VAR (GVAR)* bloc: in place of a raw foreign-variable block it carries three trade-weighted Rest-of-World “star” variables—foreign demand (ROWDEM), foreign inflation (ROWINF), and foreign policy/financial conditions (ROWFIN)—built from IMF trade weights. The US is the closed-economy anchor ($n = 36$). Conditioning any bloc propagates

to all others through the shared RoW stars, solved at display time as the linear cross-country fixed point $\Delta = M\Delta + d$; a Gauss–Seidel sweep and a direct solve agree to $\sim 5 \times 10^{-11}$ (spectral radius $\rho(M) \approx 0.59$, ~ 10 sweeps). All six models are estimated through 2025Q4. The China bloc uses a tighter Minnesota tightness ($\lambda = 0.04$).

7 User Interface Guide

7.1 Top Bar

The top bar provides global controls:

- **Country selector:** Switch between US, EA, UK, JP, CA. Switching reloads all BVAR data and resets conditioning.
- **User Manual:** Download this document as a PDF.
- **Uncertainty mode:** “None” (deterministic), “Quick” ($D = 50$ draws), or “Full” ($D = 200$ draws).
- **Compute button:** Triggers the simulation smoother (Section 4.6).

7.2 Chart Panel

The main panel displays a grid of time-series charts. The first row shows the primary macroeconomic triad—inflation, policy rate, GDP growth—plus government spending. Subsequent rows show core inflation, unemployment, consumption, investment, and any country-specific variables.

Each chart includes:

- A **dotted baseline** forecast with 68% credible bands (shaded).
- A **solid scenario line** showing the conditional forecast.
- A **dashed counterfactual line** (when active) showing the policy counterfactual.
- A **“Condition” button** to toggle conditioning on the variable.
- **Drag interaction:** Click and drag data points on conditioned charts to set target paths. Right-click to toggle individual quarters on or off.

7.3 Control Panel (Sidebar)

7.3.1 Scenario Section

- **Scenario name:** Editable text field.
- **Reset:** Clear all conditioning to return to baseline.

- **Save/Load:** Export and import scenarios as Excel files. The Excel format includes sheets for settings, conditioning paths, active quarters, rate adjustments, and a variable map for reproducibility.
- **Example:** Download a pre-built inflation-surge scenario.
- **Conditioning sliders:** Per-variable, per-quarter deviation sliders (± 3 pp) for each conditioned variable.

7.3.2 Policy Counterfactual Section

- **Counterfactual toggle:** Enter/exit counterfactual mode.
- **DSGE model browser:** Select from 169 models, searchable by name or reference.
- **Policy rule selector:** Choose from 9 standardized monetary policy rules (Table in Section 5.1).
- **Rule-based / Optimal toggle:** Switch between specifying the rate path via a Taylor-type rule or via the [Barnichon and Mesters \(2023\)](#) optimal policy (Section 5.4).
- **Dual-mandate slider:** In optimal mode, adjust the relative weight on price stability versus full employment.

7.4 Export

The scenario bar at the bottom provides four export formats:

- **CSV:** Tabular data with columns for each variable’s Q16 (16th percentile), baseline (median), Q84 (84th percentile), active scenario, deviation from baseline, and counterfactual (if active). Rows span historical quarters through the full forecast horizon.
- **Excel:** A workbook with per-variable detail sheets (Q16, baseline, Q84, scenario, deviation, counterfactual) and a summary sheet showing all variables side by side.
- **PNG:** A ZIP archive of high-resolution chart images (1200×500 px) capturing the full visual state including all rendered scenarios, uncertainty bands, and counterfactuals.
- **PDF:** A multi-page PDF with one chart per page (landscape orientation), including metadata footers with country, timestamp, and chart title.

All tabular exports (CSV, Excel) respect the active growth-rate toggle: if year-over-year is selected, the exported values reflect the y/y transformation. Visual exports (PNG, PDF) capture exactly what is displayed on screen, including all visible scenarios and their uncertainty bands.

7.4.1 Multi-Scenario Support

The tool supports up to four simultaneous scenarios. Each scenario’s conditional forecast and uncertainty bands are cached when computed, so that switching between scenarios preserves their results. Tabular exports include data for the active scenario; visual exports capture all visible scenario lines and bands.

8 Technical Notes

8.1 Client-Side Computation

All matrix algebra, Kalman filtering, and simulation smoothing are performed in the browser using TypeScript with `Float64Array` storage. The linear algebra kernel implements:

- LU decomposition with partial pivoting for F_t^{-1} (singularity threshold: 10^{-15}).
- Cholesky decomposition for shock draws ($\Sigma = LL'$; diagonal check threshold: 10^{-14}).
- Row-major storage for cache-friendly matrix operations.

8.2 Performance Optimizations

- The unconditional smoother result is cached by a fingerprint of the model dimensions and a sample of coefficient values, avoiding recomputation when only the conditioning paths change.
- Conditioning input is debounced at 30 ms to avoid recomputation during interactive drag operations.
- The 2,500 posterior draws (≈ 44 MB for the US model) are loaded lazily—only when the user clicks “Compute.”
- Plotly.js is loaded via dynamic `import()` to avoid blocking initial page render.

8.3 Numerical Conventions

- Measurement noise: $H = 10^{-12} I_n$, ensuring the Kalman filter treats conditioned values as near-exact observations while maintaining numerical stability.
- Growth-to-level conversion: $\Delta\ell = \Delta g/4$ for log-transformed variables, reflecting the annualization factor in (2).
- Tikhonov regularization for Barnichon–Mesters: $\alpha = 10^{-6}$.
- All matrix operations use IEEE 754 double precision (64-bit floating point).

9 Discussion

The BVAR Scenario Tool is designed to bridge the gap between the methodological sophistication of conditional forecasting techniques developed in the academic literature and the practical needs of policy economists who require rapid, interactive scenario analysis. We conclude with a discussion of design choices, limitations, and directions for future work.

9.1 Design Philosophy

Several design choices merit discussion. First, the three-layer architecture—baseline, scenario, counterfactual—deliberately separates concerns. The baseline forecast reflects the BVAR’s unconditional projection and is computed offline. The scenario layer allows the user to impose conditioning constraints without specifying the structural source of the deviations, following the tradition of [Waggoner and Zha \(1999\)](#). The counterfactual layer then overlays structural identification via DSGE impulse responses, enabling “what-if” policy analysis while preserving the reduced-form covariance structure of the BVAR.

Second, the decision to re-condition the counterfactual through the BVAR (rather than relying solely on the DSGE IRFs) reflects a deliberate methodological choice: the BVAR’s estimated covariance structure captures empirical co-movements that may be absent from or imperfectly captured by any single DSGE model. The IRF-consistent mode (Section 5.3c) strikes a balance by using the DSGE model to identify the responses of key variables (inflation, output) to the policy shock, while allowing the BVAR to propagate these effects to the remaining 30+ variables in a data-consistent manner.

Third, all computation runs client-side. This eliminates server infrastructure costs and latency, enables offline use, and avoids transmitting potentially sensitive forecast data over the network. The computational cost is manageable: the Kalman filter and smoother for a 33-variable, 3-lag VAR over a 20-quarter horizon completes in under 50 ms on modern hardware.

9.2 Limitations

The current implementation has several limitations that users should bear in mind:

1. **Linear, time-invariant dynamics:** The VAR coefficients are fixed at their posterior mean for the deterministic conditional forecast. Time-varying parameters, stochastic volatility, and regime switching are not supported. In periods of structural change, the fixed-coefficient assumption may be particularly restrictive.
2. **Gaussian innovations:** The Kalman filter assumes Gaussian innovations. Fat-tailed or asymmetric shock distributions, which may be important during financial crises, are not accommodated.
3. **Point-identified counterfactuals:** The DSGE IRFs used for counterfactual analysis are computed under a single monetary policy rule and model specification. The tool does not

average over model uncertainty or rule uncertainty, though the user can manually explore different model–rule combinations.

4. **Lucas critique:** The counterfactual analysis conditions on estimated reduced-form dynamics that may themselves change under the alternative policy. The IRF-consistent mode partially addresses this by using structurally identified responses for the key transmission variables, but the remaining variables’ responses are governed by the estimated BVAR covariance structure, which is held fixed.
5. **Data vintage:** The BVAR is estimated on a fixed data vintage. Real-time data revisions and the publication lag of macroeconomic series are not handled.

9.3 Directions for Future Work

Several extensions are planned or under consideration:

- **Time-varying parameters:** Incorporating stochastic volatility or time-varying coefficients (e.g., [Sims and Zha, 1998](#)) to capture evolving dynamics.
- **Structural identification:** Extending the counterfactual layer to support sign restrictions, narrative restrictions, or scenario priors ([Farkas and Jakab, 2026](#)) for identification within the BVAR itself, rather than relying on external DSGE models.
- **Mixed-frequency data:** Supporting monthly–quarterly mixed-frequency estimation to incorporate higher-frequency indicators.
- **Model averaging:** Bayesian model averaging over DSGE specifications and policy rules for the counterfactual layer.

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